JEPA-x: Cross-Predictive Physics Grounding
for Forecastable Latent Dynamics

Kehan Wen Ziming Li Siyuan Luo Fan Shi

Department of Electrical and Computer Engineering, National University of Singapore

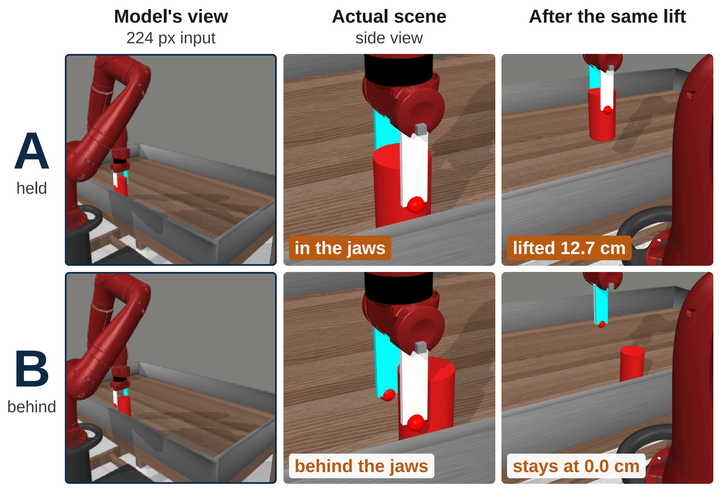
Human-centered Robotic Lab · NUS National University of Singapore · CoRL

Demos + paper
wkh923.github.io/jepa-x

1 Problem

A visual world model must predict how a scene evolves from pixels alone. **Two scenes can look almost identical to the camera yet behave differently under the same action.**

Illustrative scene



Model's views of A and B: RMS pixel difference 3.67 / 255

Before the lift, the pixels barely differ, but the physical state (contact) already does. Self-prediction from pixels has little signal to separate A from B.

Idea

1. **Pair** — Simulators give paired **camera** and **physical-state** trajectories at training time.
2. **Cross-predict** — One shared predictor forecasts **both** futures from **either** view, so the visual latent must carry contact and pose.
3. **Drop** — At test time the physical branch is removed: pixels only, same planner as the baseline.

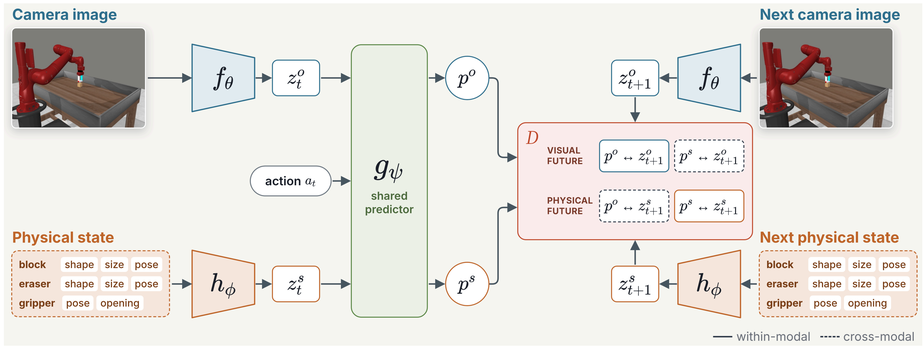
Takeaways

- Cross-modal prediction makes visual latents **more forecastable**, not just more decodable.
- Forecastable latents give **better planning from pixels**: 53.6 → 78.2% control.
- Physics is used **only in training**: same encoder, predictor and planner at test time.

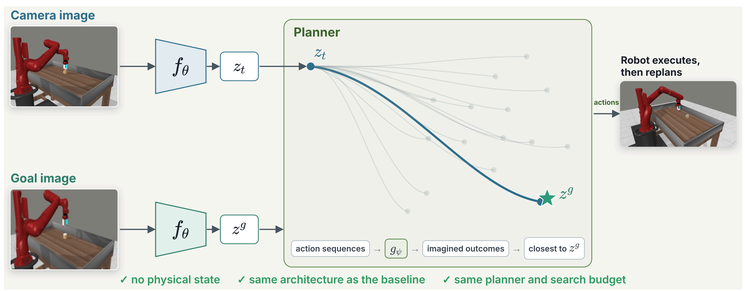
Scope. Simulation only; training needs paired physical trajectories and a rigid-body state interface.

2 Method predict both futures from either view

Training paired trajectories, both encoders online



Deploy pixels only, physical branch removed



$$\mathcal{L}_{\text{JEPA-}x} = \sum_{m,n \in \{o,s\}} \left\| g_\psi(Z_t^m, a_t) - z_{t+1}^n \right\|_2^2 + \beta \left[\Omega(z^o) + \Omega(z^s) \right]$$

m = n: within-modal (keeps LeWM self-prediction) · m ≠ n: **cross-modal** · Ω: SIGReg isotropy regularizer, β = 0.09

Planner: z-CEM over a 32-d action-VAE latent (5 macro-steps × 5 actions, 300 candidates, 30 iters), executes 25 actions, then replans. Same budget as the visual-only baseline LeWM (Maes et al., 2026).

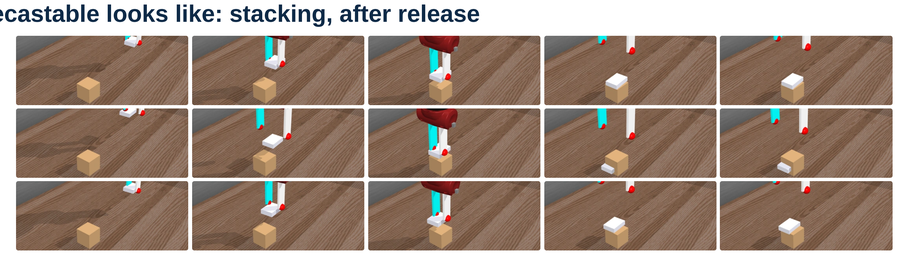
Readable ≠ forecastable: cross-prediction is what improves control

Training objective	Decodable ↑ object-position R^2	Forecastable ↓ drift, new predictor	Useful ↑ control success %
Visual-only LeWM self-prediction	0.978	0.361	53.6
Regress + state regression	**0.991**	0.373	53.7
Cross-only separate predictors	–	0.101	73.0
JEPA-x shared predictor	**0.978**	**0.104**	**78.2**

Most readable, no better control: regression makes position easiest to decode, yet the latent is no more forecastable. Cross-modal prediction is the main contributor; JEPA-x and Cross-only reach similar low drift.

Mean of 3 seeds; Distill: R^2 0.991, drift 0.516, control 57.7.

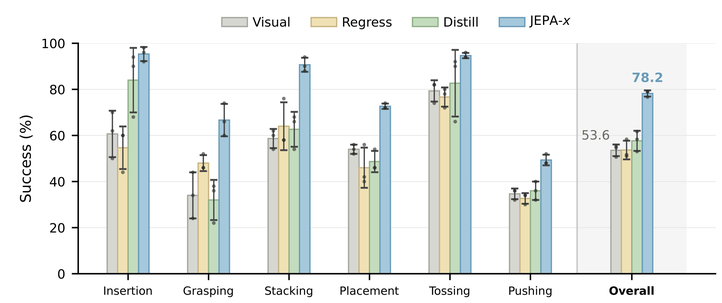
What forecastable looks like: stacking, after release



Decoded forecasts under identical actions (paper Fig. 5), not control executions. Visual-only predicts the eraser sliding off; JEPA-x keeps it supported.

3 Evidence

Control success per task family

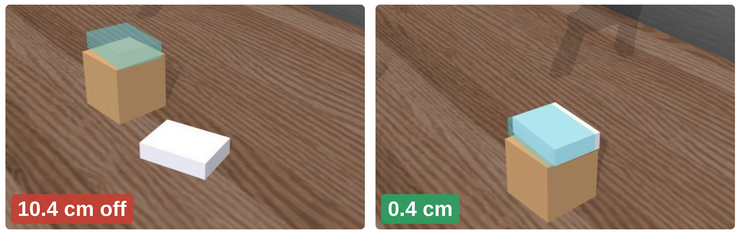


Paper Fig. 3. One model trained on 22 object–task configurations; same planner and search budget for all methods; 6 families × 50 held-out scenarios × 3 seeds. Paired sign-flip $p \leq 10^{-5}$ per seed vs. Visual. Rollout drift 0.361 → 0.104.

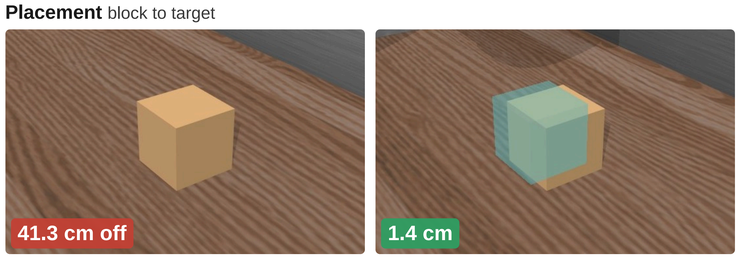
Same start, same goal selected examples

Visual-only | **JEPA-x**

Stacking eraser on block



Placement block to target



Number = final object-to-goal distance in that demo episode.

Is it really the physics?

Shuffled pairs	mismatched visual–physical pairs: control **3.6%**
Future-state aux.	predict future state from pixels: 51.3 / 54.4% (≈ baseline)
Rank-matched	at equal effective rank (57): drift **0.121** vs 0.453
Single-task	lowest drift in all 4 envs; control +20.2 Two-Room, +13.8 OGBench-Block, no gain on Push-T